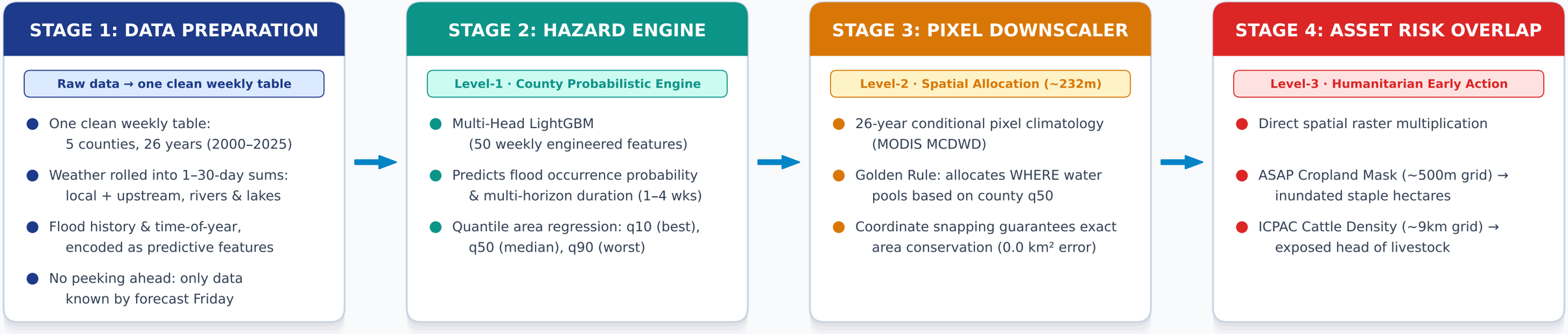
STAGE 1: DATA PREPARATION
Raw data → one clean weekly table
One clean weekly table: 5 counties, 26 years (2000–2025)
Weather rolled into 1–30-day sums: local + upstream, rivers & lakes
Flood history & time-of-year, encoded as predictive features
No peeking ahead: only data known by forecast Friday
STAGE 2: HAZARD ENGINE
Level-1 · County Probabilistic Engine
Multi-Head LightGBM (50 weekly engineered features)
Predicts flood occurrence probability & multi-horizon duration (1–4 wks)
Quantile area regression: q10 (best), q50 (median), q90 (worst)
STAGE 3: PIXEL DOWNSCALER
Level-2 · Spatial Allocation (~232m)
26-year conditional pixel climatology (MODIS MCDWD)
Golden Rule: allocates WHERE water pools based on county q50
Coordinate snapping guarantees exact area conservation (0.0 km² error)
STAGE 4: ASSET RISK OVERLAP
Level-3 · Humanitarian Early Action
Direct spatial raster multiplication
ASAP Cropland Mask (~500m grid) → inundated staple hectares
ICPAC Cattle Density (~9km grid) → exposed head of livestock